

Algebraic Thinking Task Implementation and Reflection
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The "Staircase" task provided a valuable opportunity to elicit and analyze algebraic thinking from two individuals, my father-in-law (FIL) and mother-in-law (MIL), as they engaged with a modified version of this task on Desmos Polypad, where they determined the number of cubes for increasing staircase steps, culminating in an n -step generalization, with their performance subsequently rated using the Mathematics Understanding Rubric (Mathematics Understanding Rubric.docx) across procedural fluency, conceptual understanding, and problem-solving, while also highlighting elicited thinking and pedagogical implications. My father-in-law's (FIL) procedural fluency was rated 3, reflecting his appropriate use of recursive addition ($1+2+3\dots$) to determine cube counts for initial steps, accurately identifying the procedure for the 50-step staircase ($1+2+\dots+50$) but explicitly stating it was "too cumbersome," thereby indicating an omission in applying an efficient strategy rather than an error in understanding the operation (Callahan, 2019). His conceptual understanding earned a rating of 3, as he demonstrated good evidence of conceptual knowledge, clearly understanding that each successive staircase added a consecutive integer, and while he initially succumbed to the "doubling a quantity" misconception, he swiftly self-corrected upon questioning and visual observation, showing a robust grasp of the pattern's quadratic growth. For problem-solving and strategic competency, FIL was rated 2, due to his appropriate initial strategy of physically building with digital cubes, implementing a recursive numerical approach, and making a reasoned decision to halt at the 50-step problem because of its perceived impracticality, yet not independently moving towards an abstract generalized solution. For FIL, the next teaching steps would involve bridging his recursive understanding to an explicit formula by guiding him through the geometric derivation of $n(n+1)/2$ to help him "abstract from computation" (Driscoll & Moyer, 2001), and engaging him in visual proof practices to reinforce conceptual understanding.

My mother-in-law (MIL), conversely, achieved a rating of 4 in procedural fluency, demonstrating extended use of number operations without errors in calculation and appropriate use of models, despite initial struggles where she proposed " $50 \times 50 / 2$ " (an area misapplication) and " $50!$ " (confusing factorial with summation); however, through targeted questioning, she exhibited remarkable self-correction, distinguishing operations and recognizing formula flaws, critically integrating procedures by working backward from the explicit $n(n+1)/2$ formula (found online) to derive its geometric justification. Her conceptual understanding also merited a rating of 4, showing a clear understanding of concepts and associated procedures, as evidenced by her early "sort of triangle" insight and her exemplary explanation of how two identical staircases geometrically combine to form an $n \times (n+1)$ rectangle, whose total cubes, when halved, represent the staircase count, thereby showcasing broad translation from one mode to another (Mathematics Understanding Rubric.docx) and "building rules to represent functions" (Driscoll & Moyer, 2001). In problem-solving and strategic competency, MIL earned a 4, attributed to her efficient and sophisticated approach, displaying remarkable persistence and strategic flexibility by cycling through various strategies,

critically evaluating erroneous scaffolding, proactively seeking external resources when her own methods faltered, and most notably, engaging in "doing and undoing" (Driscoll & Moyer, 2001) through backward reasoning to justify the formula's origin, thereby achieving solutions with connections, synthesis, or abstraction (Mathematics Understanding Rubric.docx). For MIL, future instruction would focus on formalizing and communicating her geometric proof, extending her skills to more complex patterning tasks like "Skeleton Towers" (Kara et al., 2015), and encouraging "abstracting from computation" (Driscoll & Moyer, 2001) in less visually obvious patterns.

This entire experience highlighted significant pedagogical implications for eliciting student thinking (Callahan, 2019), underscoring that anticipating misconceptions is crucial for targeted questioning, and my own flawed scaffolding revealed the critical importance of rigorous task pre-completion and precise, timely guidance, as ineffective support can hinder the elicitation process. The observation reinforced that "effective teaching of mathematics uses evidence of student thinking...to adjust instruction continually" (Module 6 Script.pdf), requiring careful listening and "attending" to student responses (ETS Eliciting Student Thinking.pdf) to interpret their reasoning and inform instructional adjustments, with the fundamental question of "How did you do that?" (Ginsburg, 1997) serving as a powerful tool in uncovering underlying thought processes.

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References

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